

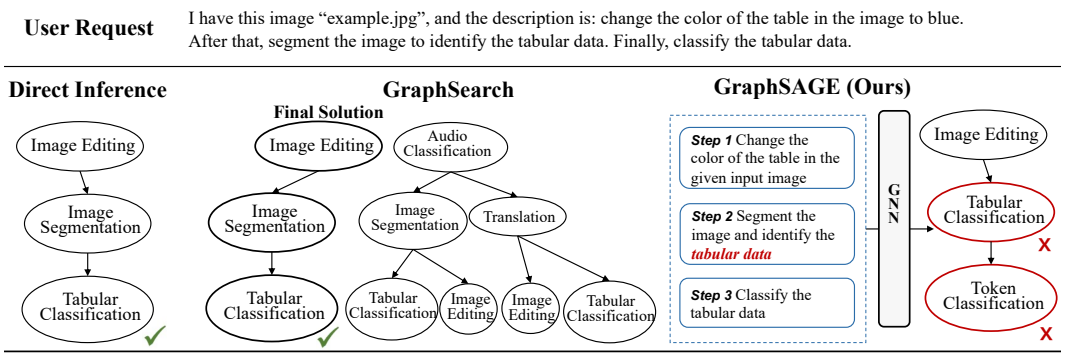


Figure 10: **Failure cases of GNN.** Our framework relies heavily on the quality of decomposed task steps. Ambiguous steps (step 2 which actually incorporates two steps) may mislead GNN to select the wrong task.

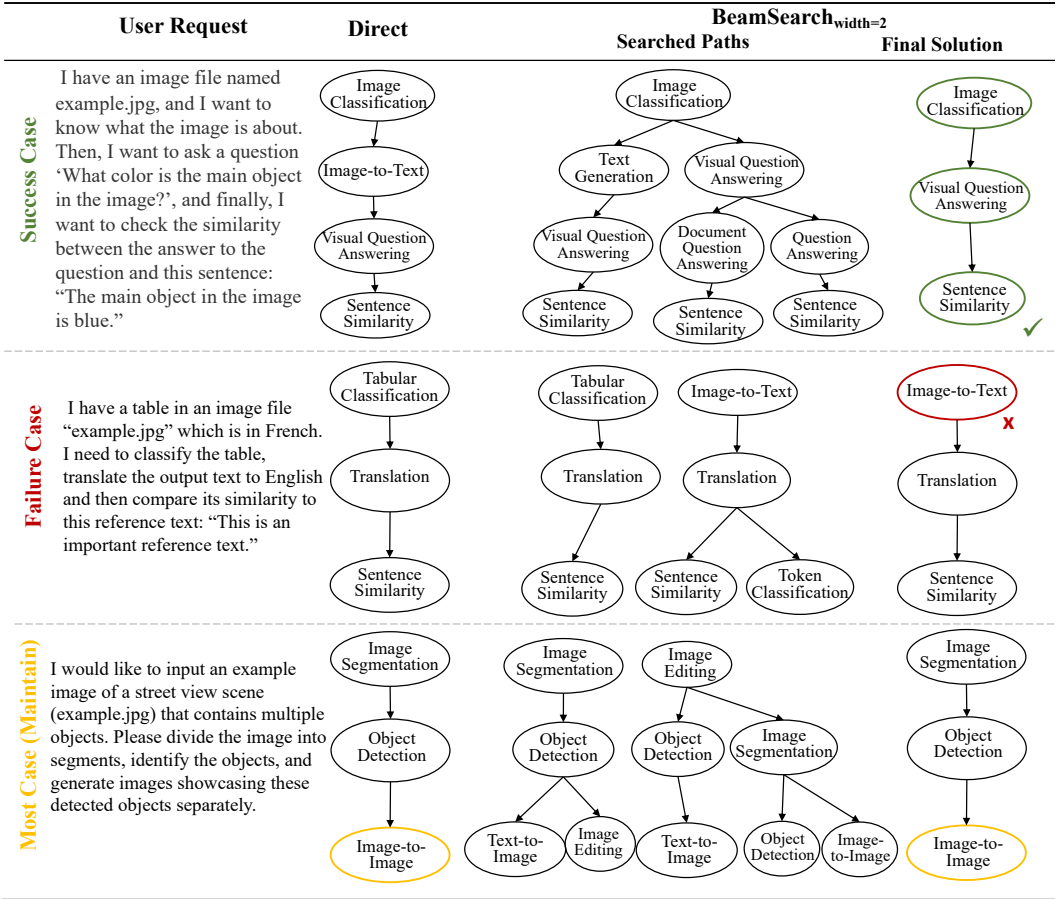


Figure 11: **Diagnosing GraphSearch for GPT-4.** We provide three types of cases: **Success Cases** where GPT-4 can leverage inherent reasoning abilities to select the optimal invocation path, which may even fix wrongly decomposed task steps. **Failure Cases** where GPT-4 miss in the extremely long context, containing the whole task graph, all searched paths, and instructions, selecting an unsatisfactory path. **Maintain Cases** where the searched result is the same as direct inference result, and those wrongly predicted tasks can still not be refined even under exhaustive search.